

Simulation study: mixture of experts model

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Data simulation:

- Sample size $n = 200$
- Dimension of the Wishart distribution $p = 2$
- Number of latent components $K = 3$
- Probabilities of subpopulations $\boldsymbol{\pi} = (0.35, 0.40, 0.25)$
- Degrees of freedom $\boldsymbol{\nu} = (8, 12, 3)$
- Scale matrices of the Wishart distribution $\Sigma_1, \Sigma_2, \Sigma_3$
- Data $S_i \sim \pi_1 \text{Wishart}(\nu_1, \Sigma_1) + \pi_2 \text{Wishart}(\nu_2, \Sigma_2) + \pi_3 \text{Wishart}(\nu_3, \Sigma_3)$

Modeling:

- Bayesian mixture of experts of two Wishart distributions

```
rm(list = ls())

library(mewishart)

set.seed(123)
n <- 200 # number of subjects
p <- 2

dat <- simData(n, p,
  pis = c(0.35, 0.40, 0.25),
  nus = c(8, 12, 3),
  Sigma = NULL # default Sigmas pre-defined in function 'simData()'
)
S_list <- dat$S
Sigma_list <- dat$Sigma_list

my_K_intial <- 3

# run a mewishart model
start_time <- Sys.time() # this has to be before 'set.seed()', otherwise non-reproducible
set.seed(123)
fit <- mewishart(S_list,
  K = my_K_intial, init_pi = c(0.45, 0.40, 0.15), mh_sigma = 1,
  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE
)

Iter    1 | LL=-1913.4 | 60.2 iter/sec | acc_rate_nu=[0.000 0.000]
Iter  500 | LL=-1839.6 | 852.1 iter/sec | acc_rate_nu=[0.038 0.150]
Iter 1000 | LL=-1839.0 | 880.7 iter/sec | acc_rate_nu=[0.042 0.135]
Iter 1500 | LL=-1842.0 | 892.5 iter/sec | acc_rate_nu=[0.035 0.135]
Iter 2000 | LL=-1840.6 | 898.9 iter/sec | acc_rate_nu=[0.039 0.132]
```

```

Iter 2500 | LL=-1841.0 | 899.8 iter/sec | acc_rate_nu=[0.036 0.128]
Iter 3000 | LL=-1839.0 | 905.4 iter/sec | acc_rate_nu=[0.037 0.133]
Iter 3500 | LL=-1841.2 | 906.2 iter/sec | acc_rate_nu=[0.039 0.132]
Iter 4000 | LL=-1840.0 | 907.9 iter/sec | acc_rate_nu=[0.037 0.130]
Iter 4500 | LL=-1842.0 | 906.6 iter/sec | acc_rate_nu=[0.037 0.131]
Iter 5000 | LL=-1837.7 | 904.1 iter/sec | acc_rate_nu=[0.037 0.131]
Iter 5500 | LL=-1841.8 | 902.4 iter/sec | acc_rate_nu=[0.039 0.131]
Iter 6000 | LL=-1837.3 | 901.2 iter/sec | acc_rate_nu=[0.039 0.130]
Iter 6500 | LL=-1831.4 | 899.2 iter/sec | acc_rate_nu=[0.039 0.127]
Iter 7000 | LL=-1830.1 | 898.1 iter/sec | acc_rate_nu=[0.040 0.123]
Iter 7500 | LL=-1829.7 | 898.8 iter/sec | acc_rate_nu=[0.039 0.118]
Iter 8000 | LL=-1828.7 | 901.1 iter/sec | acc_rate_nu=[0.040 0.116]
Iter 8500 | LL=-1832.5 | 901.1 iter/sec | acc_rate_nu=[0.039 0.112]
Iter 9000 | LL=-1829.3 | 900.7 iter/sec | acc_rate_nu=[0.039 0.110]
Iter 9500 | LL=-1837.3 | 902.5 iter/sec | acc_rate_nu=[0.039 0.108]
Iter 10000 | LL=-1831.6 | 904.7 iter/sec | acc_rate_nu=[0.039 0.105]

```

```

end_time <- Sys.time()
elapsed_time <- end_time - start_time
elapsed_time

```

Time difference of 11.06292 secs

```

# posterior summaries
colMeans(fit$pi) # estimated pi

```

```
[1] 0.1602424 0.4283326 0.4114250
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.35 0.40 0.25
```

```
apply(fit$nu, 2, mean) # estimated nu
```

```
[1] 7.434107 11.792885 5.742207
```

```
dat$nu # true nu
```

```
[1] 8 12 3
```

```

# Sum all arrays in the list and divide by the number of saved iterations
sigma_posterior_mean <- Reduce("+", fit$Sigma) / length(fit$Sigma)
# View the mean covariance matrix for Cluster 1
print(sigma_posterior_mean[, , 1])

```

```

      [,1]      [,2]
[1,] 1.8632049 -0.4261785
[2,] -0.4261785 1.8519079

```

```
print(sigma_posterior_mean[, , 2])
```

```

      [,1]      [,2]
[1,] 2.0167426 0.4495534
[2,] 0.4495534 1.4739471

```

```
print(sigma_posterior_mean[, , 3])
```

```

      [,1]      [,2]
[1,] 0.9127580 0.3541902
[2,] 0.3541902 1.1033756

```

```
as.data.frame(Sigma_list)
```

```
      X1  X2 X1.1 X2.1 X1.2 X2.2  
1 0.5 0.2  2.0  0.6  4.0  0.2  
2 0.2 0.7  0.6  1.5  0.2  3.0
```

```
(a1 <- apply(sigma_posterior_mean, MARGIN = 3, function(aa) min(sapply(Sigma_list, function(x) sum((aa
```


[1] 2.24863664 0.04622745 0.38063027

Check identifiability issue

```
# re-run the model with wrong initial weights to check identifiability issue  
set.seed(1234)  
fit2 <- mewishart(S_list,  
  K = my_K_intial, init_pi = c(0.15, 0.40, 0.45), mh_sigma = 1,  
  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE  
)
```

```
Iter    1 | LL=-1914.8 | 912.4 iter/sec | acc_rate_nu=[0.000 1.000]  
Iter   500 | LL=-1832.6 | 934.2 iter/sec | acc_rate_nu=[0.060 0.072]  
Iter  1000 | LL=-1829.8 | 923.1 iter/sec | acc_rate_nu=[0.043 0.072]  
Iter  1500 | LL=-1837.6 | 931.5 iter/sec | acc_rate_nu=[0.039 0.072]  
Iter  2000 | LL=-1833.3 | 934.8 iter/sec | acc_rate_nu=[0.041 0.067]  
Iter  2500 | LL=-1833.1 | 937.1 iter/sec | acc_rate_nu=[0.040 0.066]  
Iter  3000 | LL=-1831.1 | 934.8 iter/sec | acc_rate_nu=[0.040 0.068]  
Iter  3500 | LL=-1833.6 | 936.4 iter/sec | acc_rate_nu=[0.040 0.066]  
Iter  4000 | LL=-1831.0 | 934.6 iter/sec | acc_rate_nu=[0.040 0.066]  
Iter  4500 | LL=-1833.6 | 930.7 iter/sec | acc_rate_nu=[0.039 0.066]  
Iter  5000 | LL=-1830.8 | 925.8 iter/sec | acc_rate_nu=[0.036 0.065]  
Iter  5500 | LL=-1836.3 | 928.1 iter/sec | acc_rate_nu=[0.035 0.064]  
Iter  6000 | LL=-1835.5 | 930.2 iter/sec | acc_rate_nu=[0.035 0.062]  
Iter  6500 | LL=-1832.7 | 931.7 iter/sec | acc_rate_nu=[0.034 0.063]  
Iter  7000 | LL=-1833.7 | 931.0 iter/sec | acc_rate_nu=[0.035 0.062]  
Iter  7500 | LL=-1829.7 | 931.6 iter/sec | acc_rate_nu=[0.035 0.062]  
Iter  8000 | LL=-1831.9 | 930.4 iter/sec | acc_rate_nu=[0.035 0.062]  
Iter  8500 | LL=-1831.2 | 930.3 iter/sec | acc_rate_nu=[0.035 0.062]  
Iter  9000 | LL=-1829.6 | 929.5 iter/sec | acc_rate_nu=[0.036 0.063]  
Iter  9500 | LL=-1835.0 | 929.9 iter/sec | acc_rate_nu=[0.037 0.062]  
Iter 10000 | LL=-1828.7 | 930.4 iter/sec | acc_rate_nu=[0.037 0.062]
```

```
colMeans(fit2$pi) # estimated pi
```

```
[1] 0.2827710 0.3981268 0.3191022
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.35 0.40 0.25
```

```
apply(fit2$nu, 2, mean) # estimated nu
```

```
[1] 8.068449 12.480671 3.342674
```

```
dat$nu # true nu
```

```
[1] 8 12 3
```

```

# Sum all arrays in the list and divide by the number of saved iterations
sigma_posterior_mean2 <- Reduce("+", fit2$Sigma) / length(fit2$Sigma)
# View the mean covariance matrix for Cluster 1
print(sigma_posterior_mean2[, , 1])

```

```

      [,1]      [,2]
[1,] 0.4752475 0.242679
[2,] 0.2426790 0.709609

```

```

print(sigma_posterior_mean2[, , 2])

```

```

      [,1]      [,2]
[1,] 1.9333923 0.4449044
[2,] 0.4449044 1.4053117

```

```

print(sigma_posterior_mean2[, , 3])

```

```

      [,1]      [,2]
[1,] 2.904101 0.157671
[2,] 0.157671 2.605203

```

```

as.data.frame(Sigma_list)

```

```

      X1  X2 X1.1 X2.1 X1.2 X2.2
1 0.5 0.2  2.0  0.6  4.0  0.2
2 0.2 0.7  0.6  1.5  0.2  3.0

```